

DengueRadar

Project record and Round 1 package

Everything we have done, decided and learned, in one place

DAISI Challenge Singapore 2026 | Problem A2: Real-time outbreak forecasting

Team: Braedon (lead, platform and ML), Jingyi (weather data), Ziqi (dengue and places data, then demo), Komal (story and pitch)

Compiled 30 September 2026. Every statement reflects the project as of that date.

How to read the labels in this document

Measured: we computed it from open data. **Cited:** taken from a named source. **Designed:** our decision. **Planned:** not built yet. **Unverified:** needs a source before use.

1. The project at a glance

The idea in one sentence. DengueRadar forecasts each Singapore planning area's dengue cluster activity two weeks ahead, explains why, ranks where limited effort should go first, and gives a community group-chat admin a short, sourced, ready-to-post message.

The gap we address

NEA's public cluster map and alerts show where dengue is now. We searched and found no public tool that shows where risk is heading over the next two weeks (a search-limited finding, not proof). Research forecasts do exist, including one at planning-area level, so our claim is about the last mile: a two-week horizon, a ranking, plain-language drivers and a ready-to-post message.

Where we are

Done and verified	Not done yet
Free Edition workspace checks all passed (egress, Unity Catalog and Delta, spatial SQL, MLflow, AI/BI map, Genie, Apps)	Incremental-ingestion check (Structured Streaming, Auto Loader) has not run
Daily cluster-snapshot job live since 28 Sep (day 1: 11 clusters, 117 cases)	Per-area training panel (Ziqi, due 1 Oct) and Jingyi's weekly rainfall file
18-dataset catalogue with owners	Weekly lag chart
Burden numbers measured from the MOH weekly file	Any trained model or backtest, so no performance number exists
Monthly lag chart (temperature, humidity, rain)	The real conversation with a group-chat admin
Full design documents, three research reports, two judge-panel sessions, a design-thinking session	The slides themselves (only copy exists), team name, and the problem code check
Three-slide copy fitted to the official template	

How the idea scores today (our estimate)

Two simulated judge-panel sessions scored the idea 5.4 then 5.8 out of 10 on average. Our own reading of the official rubric puts the current concept note near 67 out of 100, and about 82 if the open evidence items are closed. These are judgement calls, not results, and shortlisting is relative to other teams.

What matters most in the next few days

- Have one real conversation with a group-chat admin and use one anonymised quote, or leave the slot out.
- Build the three slides from the copy in Section 12 and get someone outside the team to read them cold.
- Run the incremental-ingestion check and keep the daily snapshot job healthy.
- Never put an unmeasured number on a slide; state impact as targets.

2. The challenge and the rules we play by

The Databricks AI Social Impact (DAISI) Challenge invites student teams from Singapore institutes of higher learning to build data-driven solutions to local social challenges using Databricks and open data from data.gov.sg. Ten teams enter a two-week mentored sprint and three are selected at Demo Day. Databricks Free Edition is free for the challenge.

Key dates (Singapore time, 2026)

Date	Event
16 Sep	Launch at Data+AI World Tour Singapore; registration opens
24 Sep	Databricks hands-on training (completed)
6 Oct, 11:59pm	Round 1 submission closes on Devpost: a three-slide pitch as PDF (team target: 4 Oct)
7 to 9 Oct	Shortlisting; top 10 announced 9 Oct
12 to 26 Oct	Virtual kickoff 12 Oct, then the mentored build sprint
27 Oct, 4 to 7pm	Demo Day; top three selected
4 Nov	Finale at the new Databricks Singapore office

Our problem statement: A2, DengueRadar

Cited (official brief): dengue is hyperendemic in Singapore, with outbreaks affecting tens of thousands of residents. Rising temperatures and erratic rainfall extend the Aedes breeding season. NEA cluster monitoring today is largely reactive; predictive risk intelligence could let communities act 2 to 4 weeks ahead.

What to build: a near-real-time pipeline combining live dengue cluster data, weather station readings and geospatial data into a forward-looking dengue risk forecast by planning area, showing streaming or incremental ingestion from live APIs, weather feature engineering and a predictive model on an interactive map. **Target demo:** map current cluster intensity; a two-week forecast trained on weather and historical cases; explainable Low, Medium and High planning-area risk; a comparison with a recent-history-average baseline. **Stretch:** hawker-centre density or green space as proxies, threshold alerts, MLflow experiment comparisons.

Round 1 judging rubric

Criterion	Weight	What judges look for
Problem fit and social impact	30%	A clear problem and a describable benefit
Solution quality and originality	30%	A thoughtful solution, not a generic dashboard
Data and technical feasibility	25%	Named open datasets, a credible Databricks architecture, feasible in two weeks
Clarity of submission	15%	A concise, well-structured concept note or deck

Rules that shape our choices

- Round 1 is an idea submission; no working build is required. The template has exactly three slides.
- Do not upload confidential, personal or customer data, or credentials. Disclose synthetic or cached data and prototype features, and credit datasets, libraries and models.
- Free Edition is serverless and quota-limited: one 2X-Small SQL warehouse, at most five concurrent job tasks, up to three Databricks Apps that can auto-stop after 24 hours, one Lakebase project, no commercial use. Exceeding quotas can stop compute for the rest of the day.
- Cited public data is allowed, so archived cluster data may be used for training if it is credited.

3. How the idea evolved

The design changed materially twice. This section records what we thought at each stage and why we changed. Nothing was deleted; superseded records are kept in *_bmad-output/* with banners.

Date	What happened	Outcome
26 Sep	Brainstorm (fishbone, empathy map, how-might-we). We broke the problem into weather, mosquito, virus, human behaviour, environment and response lag. Insights: clusters appear 2 to 3 weeks after transmission starts; most infections are unreported (hidden iceberg); residents tend to ignore banners; fear only works when paired with one easy action.	A forecast must be paired with a clear, easy action.
27 Sep	Kickoff and first forge. First design: senior-first , with Active Ageing Centre staff as the primary user, a national weekly forecast (Level 1) and a local risk score (Level 2).	Reshaped and hardened; later superseded.
28 Sep	Three more forges (subzone risk matrix and root-cause routing; no individual identification; parks and hawker centres as core). Technical feasibility research: GO with three conditions (test outbound internet, respect quotas, NEA already forecasts). Free Edition checks all passed and the daily snapshot job went live.	A buildable design, but aimed at the wrong focus.
28 Sep	Correct-course pivot. The team realised the senior-first framing missed the brief, which asks for a per-planning-area forecast with incremental ingestion. New direction: a trained two-week forecast per planning area, explained drivers and a prescriptive action plan for residents and community leaders. Seniors stay only as a secondary extra-care flag.	Approved by Braedon; docs rewritten.
30 Sep	Round 1 judging-fit forge: no routing to a named Town Council, RC or MCST; top three drivers give ranked, deduplicated actions; a second axis routes by breeding-source tag; the action layer is rule-based and auditable.	Locked.
30 Sep	Design session (this report's main subject): H4 green-space hypothesis; a budgeted inspection ranking; a weekly action card; audience views; group-chat delivery; authority routing with a jurisdiction check; three research reports; two judge panels; design thinking; a measured burden number; the three-slide copy; repository tidy.	Design locked; evidence gaps identified.

What was dropped and why. The senior-first framing, Active Ageing Centre staff as primary users, a Senior Risk Index, "Family Ping", block-level carer lists and the exposure-times-vulnerability matrix. They answered a different question from the brief. We must not describe any of them as current.

4. The final idea

The problem in our words

NEA's public cluster map and alerts show where dengue is now. Nothing public that we found shows residents and community leaders where risk is heading over the next two weeks, so prevention (emptying containers, clearing gutters, checking drains) starts late. The Aedes life cycle can be as short as seven days (NEA, to be verified), so a two-week warning spans two breeding cycles.

The solution

- **Ingest** the live NEA cluster feed and weather data incrementally into Delta tables, building the history the feed does not keep.
- **Forecast** each planning area's cluster activity two weeks ahead with a model trained on archived cluster history, judged against simple baselines.
- **Explain** the top three drivers per area in plain words.
- **Rank** where limited effort should go first ("where do the first 8 teams go?") in two lists: most cases, and highest risk per resident.
- **Prescribe** up to three ranked, deduplicated actions per area from an auditable rule table.
- **Deliver** a short, dated, sourced message that a community group-chat admin can copy into WhatsApp or Telegram, with NEA as a comparison view.

Who uses it and what decision it changes

User	Decision	How they get it
Residents	What to check at home this week	Through the community chats their admins already run (a proposed channel)
Group-chat admins and volunteers (RC, MCST council, engaged residents)	What to post; what to check nearby	An admin kit: ready-to-post text, area picker, valid-until date
NEA (as a comparison view, not the user)	Where inspection effort should go first	A weekly export file and a data contract

What makes it different

The forecast on its own is table stakes: the brief asks for it and other teams will build it. Our difference sits in the decision and delivery layers: a budgeted ranking, drivers in plain words, actions tied to who owns them, and a message designed to be posted by a person who has to stand behind it.

What we will and will not claim

- We **will** say NEA's public cluster map and alerts describe today, and that we found no public two-week forecast tool.
- We **will not** say "NEA does not predict": NEA's institute appears to run internal national forecasts.
- We **will not** say nobody forecasts by planning area: a 2026 preprint forecasts hotspots for the 55 planning areas one week ahead.
- We **will not** claim group chats are how residents are informed as a fact; it is a proposed channel until a real admin says otherwise.
- We **will not** claim community messaging reduces dengue: the evidence is weak and mostly about mosquito counts.
- We **will not** put any performance number on a slide before it is measured.

5. Data

The dataset catalogue

Designed and cited. The full list, IDs, fields and access notes are in *DATASETS.md*. The 18 datasets, checked against the data.gov.sg API on 27 Sep 2026:

#	Dataset	Source	Used for	Tier	Owner
1	Weekly Infectious Disease Bulletin Cases (2012 to 2022)	MOH	National context and burden numbers	Core	Ziqi
2	Dengue Clusters (daily snapshot, polygons and case counts)	NEA	Live input; breeding-source tags; map	Core	Ziqi
3	Areas with High Aedes Population	NEA	Local exposure signal	Core	Ziqi
4	Residents by planning area, age, sex (Census 2020)	SingStat	Population per area; 65+ share	Should	Ziqi
5, 6	Master Plan 2019 subzone and planning-area boundaries	URA	Scoring unit and Island view	Core	Ziqi
7 to 9	Historical rainfall (2017 to 2024), air temperature, relative humidity	NEA	Weather features and lags	Core	Jingyi
10	Real-time rainfall (5-minute)	NEA	Incremental ingestion; current rain	Core	Jingyi
11, 12	NParks parks and nature reserves; hawker centres	NParks, NEA	Green-space share (H4); places where people gather	Should	Jingyi
13	Historical daily weather records	NEA	Optional: extend weather to 2012	Should	Jingyi
14 to 16	HDB property information; HDB existing building; CHAS clinics	HDB, MOH	Block age, block locations, nearest clinic	Should	Ziqi
17	Archived NEA clusters (SG Outbreak/SGCharts; Nature Scientific Data 2022)	External, cited	Training panel and the 2020 replay	Core (training)	Ziqi
18	Weather station locations	NEA	Map stations to planning areas	Core	Jingyi

Measured: how big the problem is

Measured 30 Sep by summing the MOH weekly bulletin file (*analysis/data/weekly_dengue_ziqi.csv*). This resolved a conflict between two team documents: the correct 2020 total is 35,261, not 35,012. Ziqi or Komal still confirms the source and cites it.

Year	Cases	Peak week	Note
2017	2,759	2017-W02 (90)	Lowest recent year
2019	15,910	2019-W28 (661)	
2020	35,261	2020-W30 (1,791)	53 epi weeks in the file; record year
2021	5,251	2021-W01 (194)	
2022	32,130	2022-W21 (1,563)	Second surge

The swing from 5,251 cases in 2021 to 32,130 in 2022 shows how volatile the disease is. It also shows that weather alone cannot explain surges, which shapes how carefully we word our claims.

The live snapshot job and the history gap

- NEA's cluster feed is a **today-only snapshot**: no history (*DATASETS.md*). A Lakeflow Job runs daily at 21:00 and saves each day's clusters to a Delta table, so the history exists only because we build it. It has run since 28 Sep (day 1: 11 clusters, 117 cases).
- The **training archives** (SG Outbreak/SGCharts: 256 snapshots, 3 Jul 2015 to 6 Nov 2020; Nature Scientific Data 2022: 20 weekly snapshots, 28 Feb to 9 Jul 2020, 213 subzones mapped with at least 92% of points inside the matched subzone) end in Nov 2020.
- So there is a **five-year gap** between the end of the archives and the start of the live feed. A rolling backtest cannot say the model holds in the current regime, so we treat the live snapshots as a **prospective, out-of-time test** and say so on the slides.

- 2020 has dense snapshots to May, then monthly only, and cases outside clusters are missing. We will report the weeks and areas with usable data.

Target caveat (must appear on the slides)

The archive holds only cases inside NEA clusters. The model predicts **cluster activity** (new or growing cluster cases in a planning area over the next two weeks), not total dengue cases.

A note on boundaries

Planning areas do not match Town Council, constituency or MCST boundaries, and no open dataset maps one to the other. The HDB property file has a *bldg_contract_town* code per block, which might support a block-to-HDB-town mapping, but that is not the same as Town Council boundaries and does not cover condos. It is recorded as a possible roadmap input only.

6. Method: hypotheses, models and validation

Targets

- **Regression target:** cluster cases in planning area A over weeks $t+1$ and $t+2$.
- **Classification target ("Rising"):** cases in the next two weeks are at least 20% above the area's 4-week average and above a minimum count, so tiny numbers do not flap.
- **Risk level shown to users:** Low, Medium or High from the calibrated Rising probability. Posts lead with "higher than usual" and show the level second.

The four hypotheses

Following the hackathon handbook, each hypothesis links a domain factor to a measurable feature. Any may be disproven, and we will say so.

ID	Hypothesis	Features	Status
H1 Weather lag	Warm periods, and rain 2 to 4 weeks earlier followed by dry spells, raise next-fortnight cluster activity (faster breeding, standing water)	hot_days_31c, rain_lag2..4, temp_lag, humidity_lag, flushing_days	Partly measured (monthly): max temperature +0.29 at a 4-month lag, humidity -0.26, rain -0.10. Weekly chart pending.
H2 Spatial spillover	Clusters in neighbouring areas raise an area's own risk	cluster_cases_adjacent, dist_to_nearest_cluster_km	Untested
H3 Recurrence	Areas with clusters in the past year recur, especially in older housing	cluster_weeks_last_52, median_block_age	Untested
H4 Green space (added 30 Sep)	Areas with more green space have more cluster activity, after controlling for population density . Say "associated with", not "causes"	green_space_share (NParks parks / land area), pop_density, median_block_age	Untested. Needs the leave-areas-out check, because it never varies within an area

Baselines and models

Name	Rule	Why
B1 Persistence (main)	Next two weeks equal the last four weeks' average cluster cases in that area	What a reader of NEA's dashboard would guess
B2 Seasonal average	Same week averaged over training years	Dengue is seasonal
B3 Explainable points score	Hand-set points: cluster size, high-Aedes zone, national trend, recent rain, hot days	Shows what training adds; also our honest fallback
M1 Trained model	LightGBM (regression plus a Rising classifier) logged in MLflow; a Poisson or logistic model as a readable cross-check	Handbook default for tabular data

Fallback rule: if M1 does not beat B1 and B2, we say so and present the explainable score with the honest result. The best model is registered in Unity Catalog.

How we test honestly

- **Time-ordered, never random.** Rolling origin: train 2015 to 2018, test 2019; train 2015 to 2019, test January to July 2020. Weeks are never shuffled.
- **Leave-areas-out check** (GroupKFold by planning area): confirms the model learns weather and spillover patterns, not each area's own history.
- **No leakage.** Features for week t use only data available before week t . NEA publishes cases about a week late, so we use lags of one week or more. Imputers, scalers and encoders are fitted inside each fold. Out-of-fold predictions are logged to MLflow.
- **Prospective test.** Log each forecast with its date and score it against the live snapshots two weeks later; report how many forecast dates it covers and never blend it into the backtest numbers.
- **Known weakness, stated first:** published Singapore models catch the timing of peaks but underpredict their size.

Metrics and the cost matrix

Metric	Meaning	Target
Rising recall and precision	Of real rises, how many we flagged; of our flags, how many were real	Recall first (a missed rise costs more)
PR-AUC	Quality of the Rising probability when rises are rare	Above B1 and B3
Hit rate	Share of new or growing clusters in areas rated High	Above B1; compare with NEA's roughly 90% for its 1 km2 map
Capture rate at K = 8	Share of the next fortnight's new cluster cases that fall inside our top eight areas	Above the last-four-weeks baseline (to be measured)
Lead time (weeks)	How many weeks before an area crossed the alert size we first flagged it	Measure it; never claim it first
Alert load	Share of areas rated High in a typical week	At most 15%
MAE (cases per area per fortnight)	Average miss	Below B1 and B2

Cost matrix. A missed rising area is set at five times the cost of an unnecessary reminder, as a *starting assumption*. The Rising threshold, the minimum count, the 15% cap, the 5x ratio and the risk cut-offs are tuned on validation years and frozen before the test year is scored. The trigger for a "sustained" flag is tuned the same way; a fixed "two consecutive weeks" rule is avoided because two-week forecast windows overlap.

What the literature says (from the feasibility research)

- NEA's operational LASSO model forecasts national weekly cases 1 to 12 weeks ahead with an error of 17% at one week and 24% at three months.
- Recent cases carry most of the forecast skill; weather-only models perform poorly. Absolute humidity is the steadiest weather predictor; heavy rain can lower risk for weeks.
- Two of our early rules (hot days above 31 C and heavy rain both raise risk) conflict with parts of the literature, which is why H1 is worded carefully and why we test rather than assume.
- Training on a short weather-overlap window (about 310 weeks) argues for a simple model with few lags and rolling validation.

7. The prescriptive layer: from forecast to action

DengueRadar predicts, then prescribes. The action layer is **deliberately rule-based, not model-generated**: it is auditable and tied to NEA's public guidance, unlike unverified generated advice. All wording is a draft until checked against NEA's pages.

Two independent routing axes

Axis	What it does	Caveat
Axis 1: why it is rising (the backbone)	The top three SHAP (or rule-score) drivers map to up to three ranked, deduplicated actions; overlapping actions collapse into one line	Depends on the model, or on B3 if the model does not win
Axis 2: who caused it (bonus layer)	NEA's cluster feed tags breeding source: HOMES, PUBLIC_PLACES, CONSTRUCTION_SITES. A construction tag routes to an NEA enforcement referral; home and public-place tags feed the resident checklist	The field is sparse: 2 of 11 clusters were tagged on 27 Sep. When absent, the Area view says "root cause not yet identified by NEA". The tag is used only after a cluster exists, never as a model feature

Locked decision: no routing to a named Town Council, RC or MCST. Planning areas do not align with those boundaries and no open dataset maps them. Naming an institution would imply a routing capability the data cannot support. Every community action is a checklist that any engaged resident, RC volunteer or MCST council member in a flagged area can read, act on or escalate.

6a. Inspection priority ranking (the decision layer)

Answers "we only have effort for K areas this week, which K?". It orders areas; it does **not** claim that acting there reduces cases.

- **Two lists, both shown:** "most cases" (probability times expected cases) favours large areas; "highest risk per resident" (per 10,000 residents) favours small areas with a sudden rise.
- **Budget slider K** (5 to 15, default 8, about the 15% alert cap).
- **Metric: capture rate at K** on the backtest, against B1. Filled only from measured values.
- **Guardrails:** state "cluster cases only"; show rank change versus last week so the list does not flap; mark poor-coverage areas "low data".

6b. The weekly action card

- Area, risk band and what changed since last week.
- Top three drivers in plain words ("heavy rain 2 to 4 weeks ago, now warm"); feature names never appear.
- Up to three ranked, deduplicated actions as a tick-box checklist, each with a recommended window.
- A root-cause line from Axis 2, an extra-care flag where the share of residents aged 65 and over is in the top third, and a source link.
- A **confidence line** (backtest hit rate for that band), shown only once measured.

The action library (draft)

Eleven draft actions, each tagged to a hypothesis or axis. B-L-O-C-K and S-A-W are NEA's own prevention steps (found via search; Komal must read the source pages). Full table in Appendix A.

Steps	NEA's wording (search summary only; to verify)
B-L-O-C-K	Break up hardened soil; Lift and empty flowerpot plates; Overturn pails and wipe their rims; Change water in vases; Keep roof gutters clear and add BTI insecticide
S-A-W	Spray insecticide in dark corners; Apply insect repellent regularly; Wear long sleeves and long pants

6c. Who gets what (audience views)

One forecast, three decisions. The split is by **relevance, not secrecy**: cluster locations are already public, and a public prototype has no access control.

Lens	Decision	Sees	Does not see by default
Resident (reached mainly through a chat)	What do I do at home this week?	Own area's band, what changed, top three plain-language actions, extra-care line, adjacent areas' bands	Probabilities, island ranking, model metrics
Volunteer or group-chat admin	What should our neighbourhood check, and what do I post?	The resident view plus the community checklist and the admin kit	The full island ranking
NEA (comparison view)	Where should limited effort go first?	Both priority lists, K slider, drivers, tags, confidence and coverage, backtest scores	Resident-level content

Band	Resident	Volunteer	NEA view
Low	Routine card ("Low is not zero")	Nothing	Nothing
Medium	Card with actions	Digest	Nothing
High	Alert message	Checklist	In the ranking
Sustained or tagged	Same	Same	Flag, or referral row

Alert fatigue. Each High area is marked "new" or "persistent" (tied to H3 recurrence) so recurring areas do not cry wolf, and the backtest reports the distribution of weeks in High per area. **Language rules:** "forecast cluster activity", "prevention priority"; never "worst", "dangerous" or "safe"; every view carries the cluster-cases-only caveat.

6d. Dissemination modes and authority routing

Route by who owns the action, not by geography. National agencies need no boundary match, so they can be addressed directly; boundary-based groups are reached through the admin kit and never named. The prototype sends nothing to anyone: the app generates each audience's payload, and a person or the agency's own platform takes it from there.

Authority	Owns	Trigger	Mode
NEA	Dengue control and inspection	Weekly; flag on sustained High	Weekly export file plus a data contract
NEA enforcement	Construction-site breeding	CONSTRUCTION_SITES tag present	Structured record per trigger
NEA drain cleansing (Department of Public Cleanliness)	Drain and canal cleansing	High or rising with the rain-then-warm driver	Structured record per trigger
NParks (role unconfirmed)	Parks and nature reserves	High with high green-space share (H4)	Structured record per trigger; illustrative
Residents, RCs, MCSTs, group-chat admins	Community action	Weekly, only on a band change	Admin kit with copy buttons
CDA and MOH; HDB, JTC, SLA	Clinical response; other estates	n/a	Roadmap only

The admin kit

One post per planning area per week, only when the band changes. It carries the area name and "your estate may span a different boundary", the band and what changed, up to three actions, a source line, a date and a **valid-until** line (forwarded messages live in chats for weeks), and "a forecast, not a diagnosis". Separate WhatsApp and Telegram versions, a multi-select area picker, no bot, no token and no stored user data. Extra languages are a stretch with speaker-reviewed translations.

The data contract (v0.1, a proposal)

One versioned schema, and each authority takes only the fields it needs (Appendix B). We have no agreement with any agency; all example values are illustrative.

The jurisdiction check: what it caught

We checked our routing against official pages. Several direct page fetches failed, so most points rest on search summaries and Komal must read the pages before any slide uses them.

Our claim	Result
NEA and BCA enforce construction-site breeding	NEA is supported (Stop Work Orders, court charges, the ECO Scheme). BCA did not appear , so we now say NEA only
PUB cleans public drains	Not supported. Regular cleansing is done by NEA's Department of Public Cleanliness; PUB does structural repairs and litter devices
NParks owns park dengue control	Not found. NParks appears only in the One Health Framework and gardening advice; marked unconfirmed
MOH is the health-sector body	Partly. The Communicable Diseases Agency gives dengue clinical guidance and the 24-hour notification rule (page fetched)

8. Databricks architecture

DengueRadar on Databricks Free Edition

From reactive dashboards to a 2-week forecast with a plan of action. Tested on our Free Edition workspace on 28 Sep 2026: Unity Catalog, Delta, MLflow, AI/BI, Genie, Apps, scheduled jobs.
Not yet tested: streaming and Auto Loader for incremental weather loading (planned).

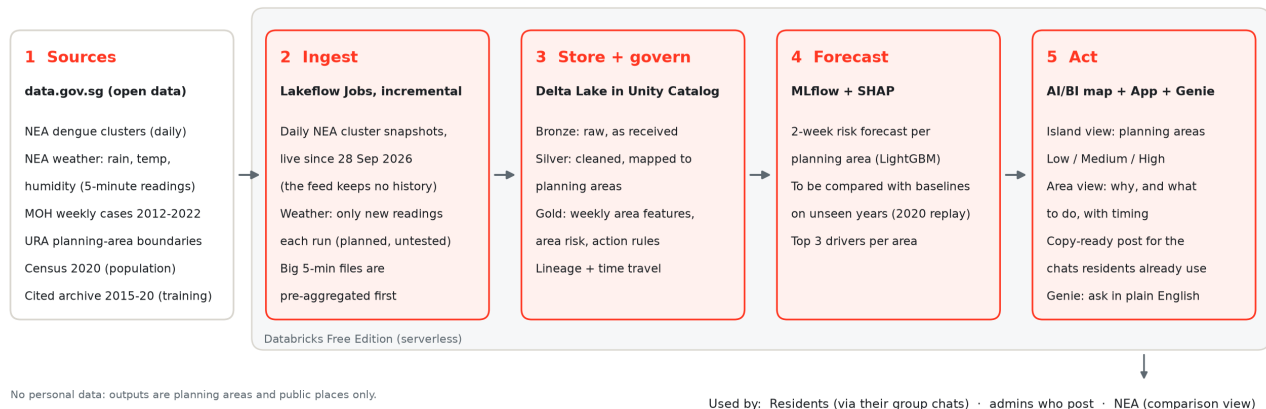


Figure 1. The architecture diagram (pitch/assets/architecture.png). The staging note on the diagram says incremental weather loading is planned and untested.

Each component and its job

Component	Its job in DengueRadar	Why not a laptop script
Lakeflow Jobs	Scheduled daily cluster snapshots (live since 28 Sep) and planned incremental weather loading	NEA's feed keeps no history; a scheduled job builds it
Delta Lake (bronze, silver, gold)	Bronze: raw as received. Silver: cleaned and mapped to planning areas. Gold: weekly area features, forecasts, action rules	Append-only snapshot history, MERGE and time travel
Unity Catalog	Governed tables, documented sources and lineage	The brief asks for cataloguing and lineage
MLflow	Compare baselines with LightGBM, register the best model, log out-of-fold predictions and SHAP drivers	Reproducible experiments and an audit trail
AI/BI map, Genie, Databricks App	Island and Area views, plain-English questions on the gold tables, the demo interface	Non-technical users read it without code

Why a platform: because NEA's cluster feed keeps no history, a scheduled Lakeflow job builds it into Delta from 28 Sep, and that same history becomes our out-of-time test. The copy-paste post is simply an output and uses no Databricks feature.

Free Edition checks (Braedon's workspace, 28 Sep)

Area	Test	Result
Egress	pypi.org; data.gov.sg dataset download, datastore API and real-time rainfall API	Yes (4 of 4)
Unity Catalog and Delta	Create schema and table; Delta history and time travel	Yes (2 of 2)
Spatial	st_dwithin in SQL; H3; shapely fallback	Yes (3 of 3)
MLflow	Log a run and model; register the model in Unity Catalog	Yes (2 of 2)
Lakeflow Jobs	Daily 21:00 schedule of the cluster-snapshot notebook	Yes; live since 28 Sep
AI/BI dashboard	Point map of today's clusters (draft Island view)	Yes
Genie	"Which cluster has the most cases?"	Yes (Taman Jurong, 80 cases)
Databricks Apps	Streamlit template starts	Yes
Incremental ingestion	Structured Streaming, Auto Loader, MERGE fallback	Not yet run

Constraints we work within

- Free Edition quotas are unpublished; keep workloads small and pre-aggregate the roughly 30 GB of raw 5-minute weather outside Databricks.
- Databricks Apps can stop after 24 hours, so a demo needs a local fallback (a cached snapshot) and a short screen recording.
- The daily snapshot job must stay healthy; the number of snapshot days is read from the table on the day, never estimated.

9. Research findings

9.1 Feasibility research (28 Sep): GO with three conditions

- **Outbound internet** is restricted on Free Edition. *Resolved*: data.gov.sg was reachable in our tests on 28 Sep.
- **Quotas are real but unpublished**. Conserve compute and pre-aggregate large weather files.
- **NEA already forecasts**. Our originality cannot rest on forecasting alone.

9.2 What already exists (competitive recon, 30 Sep)

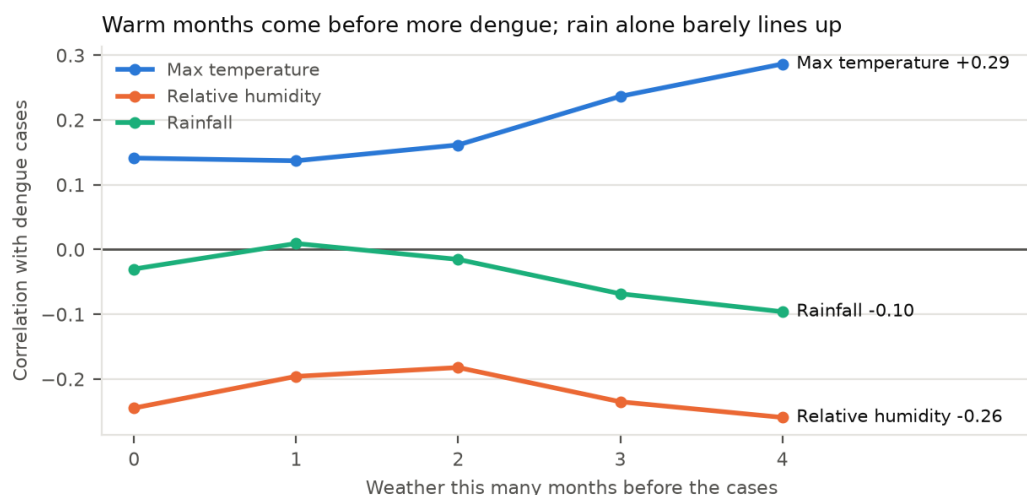
Existing thing	What it does	Forecast?
NEA cluster map and table	Active clusters, localities, case counts; Red 10+ cases, Yellow fewer than 10, Green no new cases (monitored 21 days); updated at 1am	No
NEA Dengue Community Alert System	Colour-coded banners; alerts on high Aedes mosquito population and clusters near you through the myENV app or a web portal	No forward-looking projection on the page
myENV notifications	Users save locations and enable dengue and Aedes notifications; based on Gravitrap surveillance (search summary)	Current counts, not future cases
NEA institute's internal models	LASSO models, national weekly forecasts 1 to 12 weeks ahead (search summary)	Internal
Research: neighbourhood forecast (BMC Medicine, 2018)	315 neighbourhoods, up to 12 weeks ahead, AUC above 0.75	Research
Research: hotspot forecast (arXiv preprint, 2026)	Predicts hotspots by subzone, aggregated to 55 planning areas, one week ahead, F-score 0.79; not peer reviewed	Research

Implication: keep the gap claim but narrow it ("nothing public shows where risk is heading over the next two weeks", search-limited). Position explicitly against myENV: alerts today, a forecast plus a ready-to-post message tomorrow. Cite the planning-area preprint as related work. Not searched: SG Outbreak and community projects, the myENV launch date.

9.3 Group-chat delivery and community messaging (user-voice recon, 30 Sep)

- **The channel is plausible**. WhatsApp is widely used (roughly 80 to 90% monthly, aggregator figures that conflict; treat as "widely used"). Resident-run Telegram groups exist as neighbourhood hubs with volunteer moderators. Government bodies such as REACH run chat groups.
- **Direct evidence is missing**. No source shows RC or condo chats circulating official notices, or NEA using chat groups for dengue outreach. So "residents are informed through group chats" is our team's observed practice, not a sourced fact.
- **Effect evidence is weak**. A 2025 review of 15 systematic reviews found only modest changes in mosquito indices and called the evidence weak, with most outcomes being mosquito counts rather than disease. NEA's 2022 outreach with 5,000 grassroots volunteers reports activity counts, not outcomes.
- **Risk:** forwarded messages spread fast and are often partly wrong (in a 2020 Singapore study, 52.3% of 151 adults forwarded at least one COVID message). An unofficial post carries no official authority, which is why every post has a source line, a date and a valid-until.

9.4 The lag chart: what the data says so far



Singapore 2012-2022, 132 months. Unusual weather vs unusual cases for that time of year (Spearman).
Sources: MOH Weekly Infectious Disease Bulletin, MSS monthly climate (data.gov.sg).

Figure 2. Monthly lag chart (pitch/assets/lag_chart_months.png). Singapore 2012 to 2022, 132 months, unusual weather against unusual cases for that time of year (Spearman). Measured.

Reading it: warm months come before more dengue (maximum temperature +0.29 at a four-month lag), humidity shows a negative link (-0.26) and rainfall alone barely lines up (-0.10). This supports the temperature part of H1 and argues against "more rain equals more risk". It is **monthly only**: the weekly version waits for Jingyi's rainfall file, and the line goes on a slide only if the weekly chart agrees.

10. Stress-testing the idea

We tested the idea with several methods. The judge panels are **simulated personas**, not real judges, and their scores are a structured way to find weaknesses, not a prediction.

Assumption audit, pre-mortem and Shark Tank pass (30 Sep)

Risk found	What we changed
Nothing is sent, so "delivery" was unspecified	Each audience now has a defined mode; the prototype generates payloads and sends nothing
Recurring areas would be High every week, so posts get ignored	New-versus-persistent flag; weeks-in-High report; 15% alert cap
A fixed "two weeks High" escalation rule is weak because forecast windows overlap	Tune the trigger on validation years and freeze it
"NEA would use our NEA view" is an assumption	Reframed as a comparison view, decision support beside NEA's own forecast
Too many audiences blur the story	Three lenses only; the rest on the roadmap
Benefit claims with no numbers	Impact stated as targets; benefit numbers only after the backtest
Area labels could stigmatise	Language rules: "forecast cluster activity", "prevention priority", never "worst" or "safe"

Judge panel: two sessions

A five-person panel (technical judge, business judge, sponsor judge, a group-chat admin as target user, and a demo-day sceptic) scored the idea from their own lens.

Judge	Session 1	Session 2	Main point
Priya (technical)	6	6	Cleaner and more honest; new hole: training archives end Nov 2020, live feed starts Sep 2026; nothing built
Marcus (business)	5	6	Sharp wedge; "that number is about dengue, not your idea"; no evidence anyone wants it
Aiko (sponsor)	6	6	Which parts truly need Databricks? The "history that does not exist" reason was not yet on the slide
Sam (target user)	6	6	Would try it if short, sourced, calm and multilingual; posts must not read as official; nobody has asked yet
Ravi (demo sceptic)	4	5	Round 1 needs no demo; Demo Day needs a local fallback and a recording
Average	5.4	5.8	The cleanup improved clarity and honesty but added no evidence about the product

Strengths

- Follows the brief closely and adds a real differentiator in the decision and delivery layers.
- Honest caveats: cluster cases only, a fallback score if the model does not win, and a stated data gap.
- Leak-free validation design, and a jurisdiction check that caught two errors before they reached a slide.
- The daily ingestion job is live and the Free Edition checks passed.
- The idea fits how information actually moves in Singapore neighbourhoods (to be confirmed by one real conversation).

Weaknesses and loopholes

- **No measured evidence about the product:** no backtest, no baseline number, no model.
- **No user evidence:** nobody from a group chat or RC has been asked. It is the biggest gap.
- **The five-year data gap** between the training archives and the live feed.
- **Target overreach risk:** slides must say cluster activity, not "dengue risk" for all cases.
- **Incremental ingestion is unrun**, and it is the brief's explicit ask.
- **Placeholders remain:** live cluster count, team name and problem code.
- **A bare "HIGH" could alarm a chat;** posts lead with "higher than usual".

- **Single point of failure:** the per-area panel; if it is too thin we fall back to the rule score.

Top three objections and short answers

Objection	Answer (only what is true today)
"That number is about dengue, not your idea."	It answers the template's open-data prompt. Our own evidence will be the baseline capture rate and the live-snapshot test once measured; until then we state targets.
"Your training data stops in 2020. How do you know it works in 2026?"	That is our main risk. The daily job collects live snapshots from 28 Sep and we treat them as an out-of-time test, reported honestly.
"Why does this need Databricks?"	NEA's feed keeps no history. A scheduled Lakeflow job builds it into Delta, and MLflow tracks the model against baselines.

Our rubric estimate

Criterion	Weight	Now	If open items close
Problem fit and social impact	30	about 21	about 25
Solution quality and originality	30	about 21	about 25
Data and technical feasibility	25	about 17	about 20
Clarity of submission	15	about 8	about 12
Total	100	about 67	about 82

These are our judgement, with no calibration against real judges, and shortlisting is relative to the other teams.

11. Design thinking: the admin and the resident

Goal: prepare the 15-minute real conversation and sharpen the mock post and slide 2. **We have not yet spoken to any user.** Observations are marked as sourced or as hypotheses, and the test results are deliberately blank.

Challenge statement

How might we help a community group-chat admin share a short, sourced, calm two-week dengue warning that neighbours understand and act on, without the admin having to sound like an authority or risk spreading something wrong?

Point of view

A community group-chat admin needs a post they can put their name behind, because being wrong or sounding official costs them their neighbours' trust. The difficulty is not typing the message but standing behind it.

Empathy map (hypotheses to test)

	The admin	The resident
Says	"Where is this from?" "Is it official?" "Can it be shorter?"	"Is this real?" "What do I do?" "Why is it in English?"
Thinks	"If it is wrong, people will blame me." "Will the aunties and uncles understand it?"	"Do I need to worry?" "Who sent this?"
Does	Copies and forwards from trusted sources, adds a greeting, sometimes translates, answers replies	Skims, forwards to family, ignores long posts
Feels	Responsible, cautious, wary of spamming the chat	Anxious at a bare "HIGH", numb if it repeats weekly

The journey and where an admin may drop out

Trigger, then trust check, then fit check, then edit, then post, then replies, then follow-up. The likeliest drop-outs are the trust check (no clear source), the fit check (an estate spanning two areas), the replies (no ready answers) and the follow-up (weekly fatigue).

Top concepts

Concept	Where it goes
The trust-first post: five lines at most, plain words, calm "higher than usual", three actions, a date and valid-until, and an honest source line	Slide 2 mock
The answer card: reply-ready one-liners for "Is this official?", "What do I do?", "Should I worry?"	Test in the conversation
Preview and pre-post check: show the admin what neighbours will see and confirm area and date	Sprint concept

The two prototypes we will show (illustrative, not a real forecast)

Variant A (full)	Variant B (short)
DENGUE FORECAST [Area] week of 6 Oct Next 2 weeks: higher than usual (level HIGH, up from Medium) Why: heavy rain 2 to 4 weeks ago, now warm; cases rising nearby. This week: 1. Lift and empty flowerpot plates, overturn pails and wipe rims, change vase water 2. Keep roof gutters clear 3. Use repellent, wear long sleeves and pants Valid until 19 Oct. Forecast by DengueRadar, a student project using NEA data. Not a diagnosis. Your estate may span another area.	[Area]: dengue risk higher than usual for the next 2 weeks. This week: empty flowerpot plates and pails, keep gutters clear, use repellent. Valid until 19 Oct. Source: NEA data, DengueRadar (student project).

Test plan and assumptions

Assumption	What would disprove it
An admin will post an unofficial forecast if the source is clear	The admin says they only forward official notices
"Student project using NEA data" is acceptable	The admin says it makes the post unusable
Recipients understand "higher than usual"	They read it as "someone is ill" or "nothing to do"
Recipients know what to do after reading	They cannot name one action
Admins would post weekly	They say monthly, or only when serious
One planning area fits the admin's estate	Their estate spans two or more areas

The full script, decision rules and answer card are in the outline (Section 12). If the admin refuses an unofficial source, the slide says endorsement is on the roadmap; if recipients misread the wording, we change it before it reaches a slide.

12. The Round 1 package

Round 1 is a three-slide PDF answering the official template's prompts. The template is a fixed list of one-line prompts with the right half of each slide free, so every answer must be short and each slide gets one visual. The editable template is at [pitch/daisi-round1-3-slide-template.pptx](#). Brackets need a source before submission.

Slide	Visual for the free half
1. Problem and why it matters	One large number callout: 35,261 dengue cases in 2020, peaking at 1,791 in one week
2. Solution and data	The mock post as a phone screen
3. Databricks architecture and impact	The architecture diagram

Slide 1: Problem and why it matters

Template prompt	Draft answer
Team name	[to decide]
Problem statement	A2, DengueRadar: real-time outbreak forecasting [confirm the code and wording; the template's example reads "A2 - Ageing in place"]
The problem in one sentence	NEA's public cluster map and alerts show where dengue is now; nothing public shows where risk is heading over the next two weeks, so prevention starts late.
Who is affected, and how badly	35,261 dengue cases in 2020, peaking at 1,791 in a single week, and 32,130 in 2022 (MOH weekly bulletin, data.gov.sg), plus [N] active NEA clusters today (read from the live feed).
Why it matters now	Warmer, wetter weather extends the Aedes season [source], and the mosquito's life cycle can be as short as seven days (NEA, to verify), so a two-week warning covers two breeding cycles.

Slide 2: Solution and data

Template prompt	Draft answer
Your solution in one sentence	DengueRadar forecasts dengue risk for every planning area two weeks ahead, and lets a group-chat admin post what is rising, why, and what to do, in one tap.
Who uses it, and what decision it changes	Residents, through the community chats their admins already run (a proposed channel), decide what to check at home this week. Admins decide what to post. NEA can compare a ranking of where the first 8 teams should go.
Datasets (three lines)	1. NEA Dengue Clusters, daily (data.gov.sg), plus cited 2015 to 2020 cluster archives for training. 2. NEA real-time and historical rainfall, air temperature and humidity (data.gov.sg). 3. MOH weekly dengue bulletin, URA planning-area boundaries, Census 2020, NParks parks (data.gov.sg).
What makes it different from a generic dashboard	NEA's cluster map and myENV alerts describe today. We look ahead: a two-week forecast per area, a ranking of where limited effort goes first, the drivers explained, and a ready-to-post message for the chats residents already use. Four hypotheses tested.

Slide 3: Databricks architecture and impact

Template prompt	Draft answer
Pipeline	data.gov.sg and NEA APIs to Lakeflow Jobs (daily and incremental) to Delta bronze, silver, gold in Unity Catalog to MLflow (baselines against LightGBM, SHAP drivers) to AI/BI map, Genie and a Databricks App. Why a platform: NEA's cluster feed keeps no history, so a scheduled job builds it from 28 Sep, and that history becomes our out-of-time test.
The output you will demo	Island map of Low, Medium, High per area, then click an area for the action card and a copy-ready post, plus a 2020 replay.
Measurable impact if it worked (targets, not claims)	Capture rate at $K = 8$ against the last-four-weeks baseline; lead time in weeks on the 2020 replay; alert load at most 15% of areas High; a prospective check against live snapshots (training archives end Nov 2020).
What you can realistically finish in two weeks	The pipeline, the forecast with SHAP, the ranking, the copy-ready post, the Island and Area views and the 2020 replay. Not in scope: sending to real people, agency agreements, effect sizes for actions.

Claim guardrails

Say	Do not say
"NEA's public cluster map and alerts show where dengue is now"	"NEA does not predict"
"Nothing public shows where risk is heading over the next two weeks" (we found none)	"Nobody forecasts by planning area"
"Group chats are a proposed delivery channel"	"Residents are informed through RC chats" as a fact; the 80% WhatsApp figure
Targets for the impact line	That community messaging reduces dengue, or any unmeasured performance number

The one real conversation

One 15-minute talk with a group-chat admin, an RC volunteer or an MCST council member (a relative or neighbour who runs a chat counts). No names, numbers or contact details are recorded, and permission is asked to quote anonymously. Questions: how often do you post notices and in which languages; what would you need before posting a dengue warning; would you post it and what would stop you; do official notices already reach your chat, and have you seen NEA's alerts in myENV; here is a draft, what would you change; may we quote one sentence. **If the conversation does not happen, the slot is left out. A quote is never paraphrased or invented.**

What stays off the three slides

The audience matrix, escalation tables, agency routing and jurisdiction check, data contract, action library, new-versus-persistent flag, extra languages and the feedback loop. At most one roadmap line: "next: multilingual posts and an NEA feed".

13. Plan, open items and next steps

Team roles

Person	Role	Round 1 responsibilities
Braedon	Team lead, platform and data architect, ML modeller	Workspace and Free Edition checks, incremental-ingestion check, architecture diagram, lag chart, hypotheses, baselines and metrics, the daily job, action rules
Jingyi	Weather and environment data	Weather datasets, weekly national rainfall and temperature file, station-to-area map with gaps; owns all data collection from the sprint
Ziqi	Dengue and places data, then demo	Per-area training panel with coverage table, reconciled 2020 count, post-2022 case sources; then the Island and Area views
Komal	Story and pitch	Sourced statistics, resident story, mock-ups, slides, PDF and Devpost submission

Timeline to submission

When	Step	Who
30 Sep to 1 Oct	Lock the copy; pick the team name; confirm the problem code; burden numbers with sources; read the NEA source pages; Ziqi's panel and coverage table; go/adjust checkpoint on 1 Oct	Everyone
1 to 2 Oct	Weekly lag chart (needs Jingyi's file); optional baseline capture rate; the real conversation; check the architecture diagram matches the final design (it already shows the group-chat post)	Braedon, Ziqi, anyone with an admin contact
2 Oct	Make the three visuals; build the slides in the official PPTX and export a PDF	Komal, Braedon
3 Oct	Mock judging and cold-judge read by someone outside the team; fixes	Everyone
4 Oct	Final PDF, member details, Devpost submission, confirmation screenshot (team target)	Komal
6 Oct, 11:59pm	Hard deadline	

Open items

Item	Owner	Why it matters
The real conversation with a group-chat admin	Anyone with a contact	Closes the biggest evidence gap; otherwise the slot is removed
Read the NEA pages behind B-L-O-C-K, S-A-W, the seven-day life cycle and the alert system	Komal	Most links rest on search summaries
Confirm the problem code and the file source for the case numbers	Komal, Ziqi	The template example says "A2 - Ageing in place"
Run the incremental-ingestion check	Braedon	The brief's explicit ask; slide 3 says "planned" until it passes
Per-area training panel and weekly rainfall file	Ziqi, Jingyi	Go/adjust decision on the trained model
Team name	Everyone	First line of slide 1
Re-check the diagram after final decisions (add the weekly export if wanted)	Braedon	The older PDFs predate the latest design and sit in docs/archive

If we are shortlisted (sprint, 12 to 26 Oct)

Date	Milestone
9 to 11 Oct	Turn the idea into a spec and stories with the MoSCoW list as the cut line; agree table designs
12 Oct	Kickoff with the mentor. Ask who would act on a two-week area forecast and what would make the action list credible
15 Oct	Walking skeleton: real data flows load, a baseline forecast and a rough map
19 Oct	All MUST items working; midpoint check; cut what is behind
20 to 24 Oct	SHOULD items: 2020 replay polish, extra features
25 Oct	Feature freeze, code review, record the 30-second backup video
27 Oct	Demo Day

MUST: daily archive running; incremental weather ingestion; per-area training panel; a trained two-week forecast beating the persistence and seasonal baselines in MLflow on unseen years; SHAP drivers; the action table; Island and Area views; measured lead time and hit rate. **SHOULD:** national MOH forecast as context; the 2020 replay screen; a Genie space; the budgeted ranking and admin kit; parks and hawker centres if validation keeps them. **Sprint-only builds** (scheduled after Round 1): the two gold tables, the K and threshold sliders, the audience selector, the admin kit, the agency records, and the demo fallbacks.

14. The repository

Where things live

Path	What is in it
README.md	Start-here overview and reading order
project-context.md	Challenge brief, dates, our angle, build path, judging criteria
DATASETS.md	Every dataset: source, tier, owner, how to pull it
docs/	Model definitions, action library, data contract, team plan, ML explainer, checkpoint notes; this report; docs/archive/ for superseded PDFs
pitch/	Round 1 outline and slide copy, the official template (PDF and PPTX), architecture diagram source and images
analysis/	Analysis scripts; analysis/data/ is working data and is gitignored (about 69 MB)
notebooks/	Free Edition checks and the daily cluster-snapshot job
_bmad-output/	Brainstorm, forged ideas (history), the pivot, research reports, design thinking, judge-panel memory, with an index
github/, .toolkit/	Reference playbooks (not our code); .toolkit is a subset used by the BMad party-mode configuration
_bmad/, .claude/, .agent/, .agents/, .opencode/	BMad and assistant tooling; do not edit by hand

What the tidy did (30 Sep)

- Moved the two superseded PDFs (the 28 Sep meeting pack and the earlier concept reference) into docs/archive/; nothing was deleted.
- Moved the official template PDF into pitch/ beside the PPTX and fixed the one reference to it.
- Removed a regenerable Python cache folder.
- Added an index to docs/ and to _bmad-output/, and rewrote the root README so it points to this report first.
- Extended .gitignore for OS and notebook clutter.
- Checked every relative Markdown link in our own documents: none were broken.
- Left github/ and .toolkit/ alone: they overlap, but other documents and the BMad configuration point at them.

Known leftovers: the archived PDFs and the dataset-analysis PDF still use the older framing; the sprint-change proposal is history and records the pivot; most of the project files are untracked in git and nothing has been committed in this session.

Appendix A: Action library (draft v0.1)

Wording and source links are drafts. NEA links were found by search and the pages could not all be fetched, so Komal must read each one. Rows saying CONFIRM have no source. No row names a Town Council, RC or MCST.

ID	Driver tags	Resident text	Community checklist	Days	Owner (confirm)
A01	H1	Lift and empty flower-pot plates, overturn pails and wipe rims, change water in vases	Remind neighbours to do the same on balconies and corridors	7	Residents
A02	H1	Keep roof gutters clear and add BTI insecticide as NEA advises (gully traps not confirmed)	Check and clear roof gutters and common-area standing water; report blocked public drains	7	Residents; NEA (drain cleansing)
A03	H2 cluster nearby	Apply repellent, wear long sleeves and pants, spray insecticide in dark corners; see a doctor early for fever (not confirmed in sources found)	Share the repellent and fever reminder with nearby neighbours	3	Residents
A04	H2 cluster nearby	Check your home and corridor for standing water today	Remind neighbours in the blocks around the cluster	3	Community volunteers
A05	H2 neighbour rising	Do your usual weekly checks and stay alert	Pre-check known problem spots; watch the neighbouring forecast	7	Community volunteers
A06	H3	Keep up your routine weekly checks	Scheduled walk-through of spots that had breeding before	14	Residents
A07	H3	(none)	Walk known recurring spots and note standing water	14	Community volunteers
A08	H4	Use repellent before outdoor meals or exercise; check plant pot plates	(none)	7	Residents
A09	H4	(none)	Check drains, bins and stagnant water in nearby parks and eating areas; report to NEA	7	NEA; NParks (unconfirmed)
A10	Axis 2: CONSTRUCTION_SITES	(none)	Enforcement referral: locality, case size and date go to NEA	3	NEA (BCA role not found)
A11	Extra-care flag	Check on older neighbours and family members	Check on older neighbours	7	Residents

Appendix B: Data contract (draft v0.1)

One row per planning area per week. Each authority takes only the fields it needs. This is a proposal: we have no agreement with any agency and the prototype sends nothing.

Field	Type	Meaning
contract_version	string	Schema version, "0.1"
area_id, area_name	string	URA planning area code and name (55 areas)
week	date	Monday of the forecast week
band, band_change, persistence	enums	Low, Medium, High; up, same, down, new; new or persistent
top_drivers, action_ids	arrays	Up to three plain-words drivers; up to three action IDs, deduplicated and ranked
rank_cases, rank_rate	int	Priority ranks (NEA view only)
confidence	string or null	Backtest hit rate for the band; null until measured
coverage_flag, tag	enums	ok or low_data; NEA breeding-source tag or null
caveat, valid_until	string, date	"Forecast cluster activity, not a diagnosis. Cluster cases only."; end of the forecast window

Consumer	Fields	Trigger
NEA weekly export	All fields (CSV or GeoJSON)	Weekly
Enforcement, drain-cleansing and NParks records	area, week, band, tag, drivers, actions, caveat, valid_until (JSON)	Only when the trigger fires
Admin kit post	area_name, band, band_change, drivers, actions, caveat, valid_until (text)	Weekly, only on a band change

Appendix C: Key sources

Links found or fetched on 30 Sep 2026. Where a source was read only through a search summary it is marked; those need a direct read by Komal.

Source	Used for	Read
NEA cluster map: nea.gov.sg/dengue-zika/dengue/dengue-clusters	Current-state map; tiers; updated 1am	Fetched
NEA Dengue Community Alert System: nea.gov.sg/dengue-zika/dengue/dengue-community-alert-system	myENV and banner alerts; no forecast	Fetched
NEA Stop Dengue Now: nea.gov.sg/dengue-zika/stop-dengue-now	B-L-O-C-K and S-A-W; seven-day life cycle	Search summary
NEA construction sites: nea.gov.sg/our-services/pest-control/mosquito-control/mosquito-control-in-construction-sites	NEA enforcement at construction sites	Search summary
PUB drain cleansing and maintenance page	PUB does structural repairs; NEA DPC cleans drains	Search summary (fetch returned 403)
CDA dengue fever page: cda.gov.sg/professionals/diseases/dengue-fever	Dengue notifiable within 24 hours; clinical guidance	Fetched
Neighbourhood dengue forecast: BMC Medicine, 6 Aug 2018 (PMC6091171)	Related research	Fetched
Hotspot forecasting preprint: arXiv 2601.12856 (Jan 2026)	Planning-area forecast, one week ahead; not peer reviewed	Fetched
HardwareZone / Straits Times, 22 Jul 2025: BTO Telegram groups	Resident-run chats	Fetched
Singapore WhatsApp forwarding study (PMC8709420)	Forwarded-message risk	Fetched
Community mobilisation meta-review, 21 Aug 2025 (PMC12501563)	Weak effect evidence	Fetch summary
NEA and PA volunteer outreach news release, Jul 2022	Volunteers carry dengue messages	Search result
MOH Weekly Infectious Disease Bulletin (data.gov.sg dataset d_ca168b2cb763640d72c4600a68f9909e)	Burden numbers (computed)	Measured from our copy

Appendix D: Glossary

Term	Meaning
Cluster	NEA's grouping of two or more dengue cases with onset within 14 days and within 150 m (tiers: Red 10+ cases, Yellow fewer than 10, Green no new cases, monitored 21 days)
Planning area	One of 55 URA areas; our scoring and display unit
Epi week	Epidemiological week, as used in MOH's weekly bulletin
Aedes	The mosquito genus that spreads dengue
B-L-O-C-K, S-A-W	NEA's home prevention steps and personal protection steps
Baseline	A simple prediction method the model must beat
Capture rate at K	Share of the next fortnight's new cluster cases falling inside our top K areas
Prospective test	Scoring forecasts made now against data that arrives later
SHAP	A method that shows which features drove a prediction
Lakeflow, Delta, Unity Catalog, MLflow	Databricks job scheduling, table storage, governance and lineage, and experiment tracking
Bronze, silver, gold	Raw, cleaned and ready-to-use table layers
Admin kit	Ready-to-post text, an area picker and a valid-until date for a group-chat admin